

Exam Date & Time: 29-Apr-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - APRIL/MAY 2024
SUBJECT: MAT 1271/MAT_1271 - ENGINEERING MATHEMATICS - II
(PHYSICS GROUP)

ENGINEERING MATHEMATICS - II [MAT 1271-PHY]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Find the maxima and minima of the function:

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20 \quad (4)$$
- 1B) If $u = \sin(x^2 + y^2)$ where $4x^2 + 9y^2 = 25$ then by using partial derivatives find $\frac{du}{dx}$. (3)
- 1C) If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$ then prove that $xu_x + yu_y = \sin 2u$. (3)
- 2A) Find the equation of the sphere formed by the line joining points (2, -1, 4) and (-2, 2, -2) as diameter. Further, obtain the area of the circle in which the sphere is cut by the plane $2x + y - z = 3$. (4)
- 2B) Expand $f(x, y) = e^x \cos y$ about the point $(1, \frac{\pi}{4})$ upto second degree term, by Taylor's series expansion. (3)
- 2C) Evaluate $\lim_{x \rightarrow 1} \frac{x^x - x}{x - 1 - \log x}$. (3)
- 3A) Test the convergence of the following series:

$$1 + \frac{1}{2}x + \frac{1.3}{2.4}x^2 + \frac{1.3.5}{2.4.6}x^3 + \dots \quad \text{where } x > 0. \quad (4)$$
- 3B) Evaluate $\int_1^2 \int_{\frac{1}{2}}^1 xy \, dx \, dy$ by changing the order of integration. (3)
- 3C) Use double integrals to find the common area to the circles $r = a$, $r = 2a \cos \theta$ where $a > 0$. (3)

4A) Use Laplace transforms to solve $y''(t) + 4y(t) = 5t$ with $y(0) = 0$ and $y'(0) = 1$. (4)

4B) Express the function $f(t) = \begin{cases} t^2 & 0 < t < 2 \\ 4 & 2 \leq t < 4 \\ 0 & t \geq 4 \end{cases}$ (3)
in terms of unit step function and hence find its $L\{f(t)\}$.

4C) Evaluate $L^{-1}\left\{\frac{s}{(s-2)(s^2+4)}\right\}$. (3)

5A) Test the absolute convergence and conditional convergence of the series:
 $\frac{1}{\sqrt{1^3+1}} - \frac{1}{\sqrt{2^3+1}}x + \frac{1}{\sqrt{3^3+1}}x^2 - \dots$ (4)

5B) Use triple integrals to find the volume bounded by the xy -plane, the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 3$. (3)

5C) Use beta and gamma functions to evaluate:
 $\int_0^1 x^2(1 - \sqrt{x})^3 dx$ (3)

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